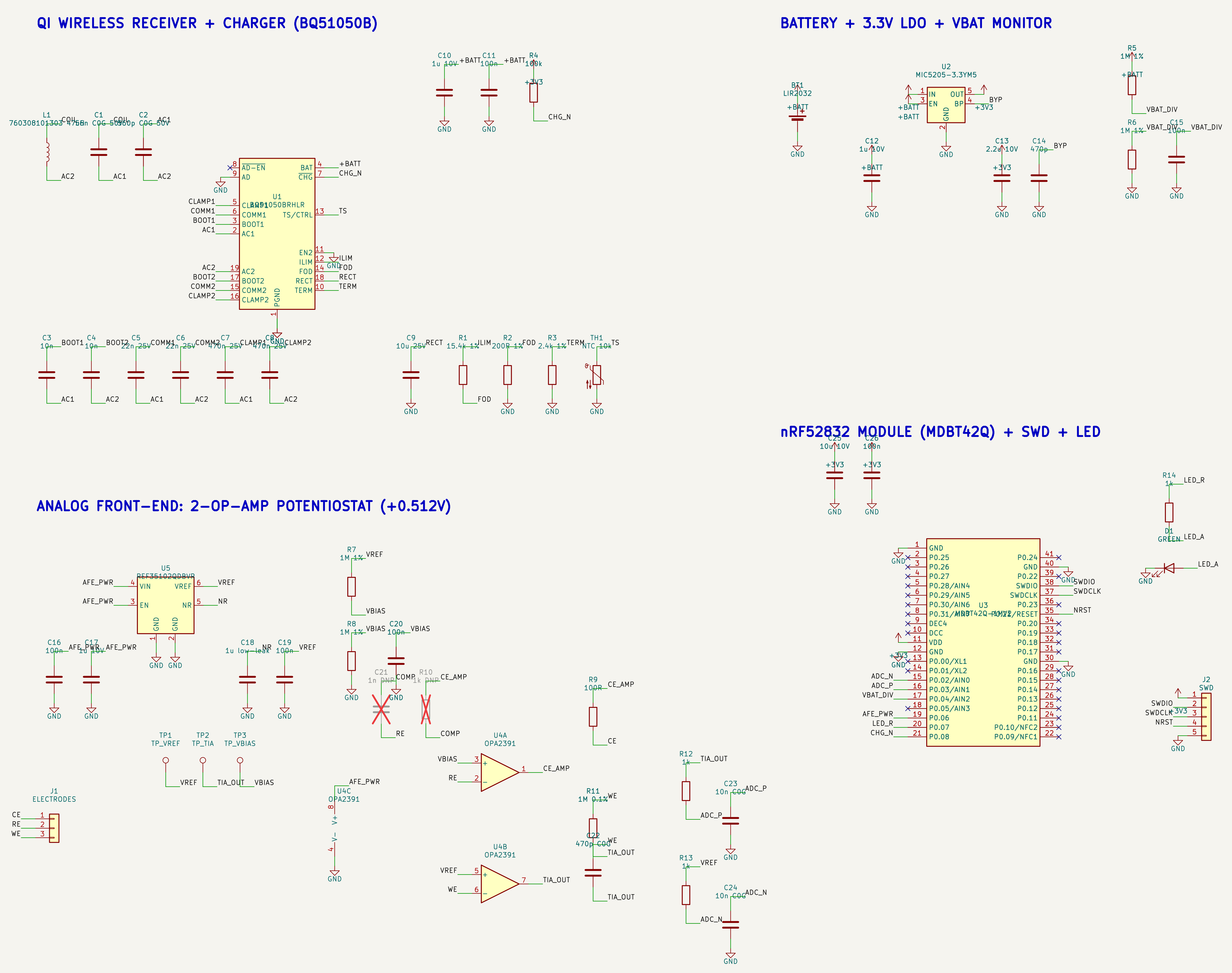

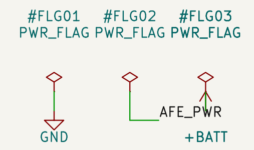

DISCRETE POTENTIOSTAT – wireless-charged 3-electrode chronoamperometry
E_cell = WE – RE = 1.024V – 0.512V = +0.512V fixed. I_WE = (V(ADC_P)–V(ADC_N))/RF

DESIGN NOTES / VERIFY BEFORE LAYOUT

1. CHARGE CURRENT: IBULK = 314/RILIM = 314/(R1+R2) = 314/15.6k = 20mA (LIR2032 0.5C). TI states BQ5105xB regulates best >=200mA; at 20mA expect loose regulation and wide termination spread (R3 2.4k = 10% term). Validate the charge profile before enclosing the cell. Safer alternative for a 40mAh coin cell: BQ51003 + BQ25100.
2. RESONANT TANK: L1 47uH (Wurth 760308101303, dia 26.3mm). C1 = 1/(L*(2pi*100kHz)^2) = 54n -> 56n C0G. C2 -> fd = 1MHz -> 560p C0G. Retune both whenever the coil changes.
3. MDBT42Q pads 22–36 are a sequential reconstruction (P0.09/NFC1 ... P0.23); pads 1–21 and 37–41 verified against the datasheet. Every pad actually used in this design is a verified one. Cross-check the full pad table against the Raytac datasheet before PCB layout.
4. TS: 10k NTC placed against the LIR2032 holder (fixed 10k = temp sense defeated, safe cond).
5. AFE_PWR is driven by nRF GPIO P0.06 set to high-drive. Load = REF35+OPA2391 ~ 60uA. Power the AFE off between runs to avoid electrode polarization.
6. WE net: guard ring driven from VREF, no soldermask opening at TIA input, clean flux.
7. RF/CF options: 1M/470p (default, +2.1uA/–0.9uA FS), 5.1M/100p (427/181nA), 10M/47p (217/92nA). rf_ohm + oversample are runtime settings in firmware; recalibrate after a swap.
8. R10/C21 control-amp compensation are DNP; fit only if the CE–RE loop oscillates.
9. LIR2032 has no protection circuit: firmware must System–OFF below 3.0V (VBAT_DIV, AIN2).
10. SAADC: differential P=AIN1(ADC_P) N=AIN0(ADC_N), gain 1/2..4, oversample 64x default.

E = +0.512V fixed; I range set by RF (default 1M)
OPA2391 + REF35102 + nRF52832(MDBT42Q) + MIC5205 + BQ51050B + LIR2032
discrete–potentiostat
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Title: Discrete Potentiostat – wireless 3–electrode chronoamperometry

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